

LLM4HIN: Discovering Meta-path with Large Language Model for Reasoning on Complex Heterogeneous Information Networks

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Abstract—Heterogeneous Information Networks (HINs) are complex structures hosting various types of entity nodes and diverse relationships. Meta-paths are crucial in HINs for relaying detailed information and enhancing knowledge transfer. However, their discovery is challenged by the expansive meta-path space inherent in complex HINs. To address this, we introduce LLM4HIN, an innovative meta-path discovery framework that utilizes Large Language Models (LLMs) for efficient meta-path discovery. Through the use of LLMs, LLM4HIN bypasses challenges including computational limitations, semantic inconsistency, and path selection optimization traditionally associated with meta-path reasoning. LLM4HIN could efficiently discovers meta-paths by utilizing LLMs to generate these paths from a minimal subset of HIN samples. Experimental evaluation on large-scale HINs attests to the enhanced performance of the meta-paths generated by LLM4HIN in link prediction tasks compared to state-of-the-art methods.

I. INTRODUCTION

Heterogeneous Information Networks (HINs), such as YAGO [1], NELL [2] and DBPedia [3], constitute intricate systems where different types of entities are interconnected through various relations. In HINs, nodes can represent various entities, including individuals, objects, organizations, and so forth, while edges denote the diverse sorts of relations, like friendship, collaborations, transactions, or influences [4]. In contemporary times, there is an increasing need to understand complex interrelationships within large and diverse networks [5], [6], such as social networks, biological networks and knowledge graphs. Researching HINs is instrumental in enhancing our understanding and simulation of the structure and dynamic behavior of these complex systems, e.g. academic

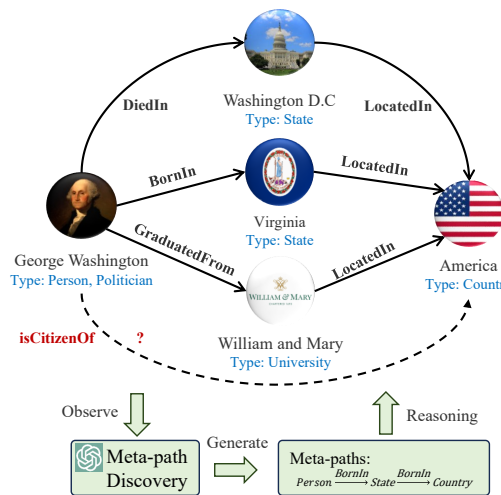


Fig. 1. An example of two-view HIN. Each entity in the instance graph is associated with an entity type in the schema graph.

events [7], commercial activities [8] and drug-target interactions [9].

To explain the detailed structural data within HINs, the concept of meta-path is introduced, affording a high-level abstraction of HINs. A meta-path is essentially a sequence of edges that connect different node types in HINs, and each edge in the sequence represents a particular type of relation. For example, as illustrated in Fig 1, the entity pair of *GeorgeWashington* and *America* can be connected through

several meta-paths, including:

$$\begin{aligned} & Person \xrightarrow{BornIn} City \xrightarrow{LocatedIn} Country \\ & Politician \xrightarrow{GraduateFrom} University \xrightarrow{LocatedIn} Country \end{aligned}$$

These meta-paths offer substantial support to the argument that *George Washington* is an *America* citizen, and could provide explanatory power for understanding the relation *isCitizenOf* among other entities and relationships.

Meta-paths are crucial for HIN reasoning tasks, such as link prediction, due to their explanatory power [10]–[12]. Additionally, meta-paths aid in knowledge transfer in inductive settings [13]. Discovering meta-path is a way to abstract the structure and capture the underlying data of HINs. Therefore, meta-path enhances prediction accuracy and provides the necessary interpretability in HIN reasoning.

While meta-path reasoning plays a crucial role in HINs, its application faces several challenges [14]. To commence, complexity [15]: the diversity of entities and relationships within HINs leads to an enormous number of potential meta-path combinations, requiring substantial computational resources for comprehensive coverage. What’s more, the ignorance of semantic information represents a significant challenge. Meta-paths carry abundant semantic information which is of great significance. Nevertheless, the majority of research, whether based on embedding or reinforcement learning (RL), ignores this information to some extent.

Large language models (LLMs), developed through the ingestion of extensive textual corpora, demonstrate a proficiency in comprehending the meaning of natural language and engaging in sophisticated reasoning that incorporates common-sense knowledge. Moreover, LLMs can perform cross-modal inference with relatively small time consumption, as LLMs are pre-trained. These two advantages endow LLMs with broad applicability and superior performance in HINs, indicating their vast potential for solving both computational efficiency and semantic ignorance.

In this paper, we propose LLM4HIN, a framework for meta-path mining that leverages LLMs, which can generate a substantial quantity of meta-paths in schema-complex HINs for reasoning and prediction with only a few examples. Compared to conventional meta-path discovery techniques, LLM4HIN demonstrates enhanced efficiency and a better grasp of semantic information. Initially, LLM4HIN acquires a subset of meta-paths from HINs which are then selected as exemplars for LLMs to produce a wider array of meta-paths. Following cleaning and validation, these generated meta-paths are utilized for link prediction.

The main contributions of this paper are as follows:

- 1) We pioneer the integration of LLMs into the task of meta-path discovery in schema-complex HINs. By harnessing the semantic understanding prowess of LLMs, we identify meta-paths that are not only of high quality but also comprehensive in nature.

- 2) By employing LLMs, the efficiency of discovering meta-paths has been significantly enhanced, obviating the need for iterative processes in the vast search space.
- 3) We do experiments based on two large-scale HINs, demonstrating that the meta-paths generated by LLM4HIN achieve superior performance in link prediction tasks compared to baselines. Then, we conduct ablation experiments to illustrate the influence of semantic information caught by LLM4HIN. After that, inductive experiments are carried out to show the generalization of LLM4HIN.

The rest of this paper is organized as follows. Related work is discussed in Section 2. In Section 3, we introduce definitions. Afterward, we describe LLM4HIN in detail. Section 4 presents results for link prediction experiments, ablations experiments, and compares LLM4HIN against state-of-the-art baseline methods on two real-world HINs. Finally, we conclude the paper in Section 5.

II. RELATED WORK

Meta-path Reasoning. The development of meta-paths by Sun et al. has led to significant research that validates its effectiveness in link prediction across multiple domains [10], [16]. The fundamental basis of meta-path reasoning approaches lies in the identification of relevant meta-paths. As mentioned in [17]. The most direct approach involves exhaustively enumerating meta-paths up to a predetermined length. However, this method proves to be computationally intensive and impractical for intricate HINs. In contrast, Lao and Cohen propose the use of random walks to construct meta-paths with a fixed length l , along with the introduction of a learnable proximity measure derived from these identified meta-paths [18]. Nevertheless, the selection of the optimal value for l significantly impacts performance and may differ substantially across various datasets. In [19], a comprehensive framework is presented for the extraction of meta-paths from complex HINs, which appropriates classical knowledge discovery methodologies and evaluates the influence of type granularity on proximity measures. Furthermore, Meng et al. introduce FSPG, a greedy strategy that leverages user input to identify the most pertinent subset of meta-paths relative to specified entity pairs [20]. Yang et al. have developed a method that concatenates deep content embedding with continuous RL to implicitly learn meta-paths, deriving similarity scores as the empirical probabilities of accessing the target entities [21]. Wan et al. in turn, have introduced MPDRL [22], a reinforcement learning framework that emphasizes type context in the generation of path instances before their summarization. However, MPDRL is vulnerable to the computational demands of its path-finding module, as its effectiveness is dependent on partial observations (i.e., the generated path instances).

Reasoning Based on LLM. Given the remarkable prowess of LLMs in natural language processing (NLP), researchers have initiated investigations into the application of these models for reasoning tasks within intricate systems such as Knowledge Graphs (KGs) [23], [24]. Although KGs are typically not

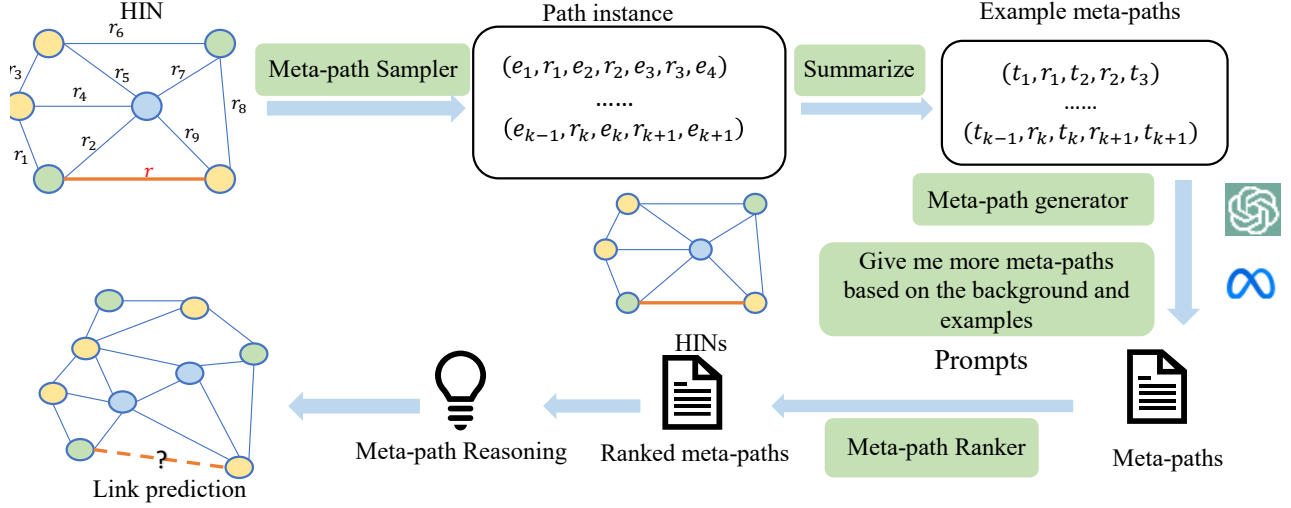


Fig. 2. Overview of LLM4HIN. Firstly, path instances are obtained by a meta-path sampler based on HIN, then summarized to derive example meta-paths. Combined with background information, LLM generates more meta-paths based on examples. After ranking, meta-paths are used to reason about facts in HINs (e.g., link prediction).

the open-world environments in which LLMs are generally trained, certain meta-paths acquired by LLMs in open-world settings can still be pertinent to these closed systems. Specifically, researchers have crafted prompts adorned with few-shot examples [25] and implemented chain-of-thought reasoning [26] to channel LLMs' reasoning within specific systems.

To augment reasoning capabilities, logical and probabilistic reasoning mechanisms have been integrated into LLMs' inference processes [27]. Furthermore, endeavors have been undertaken to infuse external knowledge into LLMs, thereby enhancing their grasp of common sense and logical reasoning [28]. In schema-complex HINs, the utilization of LLMs for reasoning tasks has not been extensively explored. We propose a novel approach to integrate LLMs into the meta-path discovery process in such networks. Our method leverages the powerful representation learning capabilities of LLMs to reason within the HIN. Our approach opens up new possibilities for utilizing LLMs in schema-complex HINs, paving the way in this domain.

III. DEFINITIONS AND NOTATIONS

Definition 1 (Heterogeneous Information Network, HIN) A HIN $\mathcal{H}$ can be defined as a directed graph $G = (V, E, \tau, \phi)$, where V represents the collection of entities within the graph and $E \subseteq V \times V$ denotes the connubium of edges connecting entities within V . The function $\tau : V \rightarrow T$ serves as the type assignment mapping, with T denoting the taxonomy of entity types, while $\phi : E \rightarrow R$ acts as the relation mapping function, where R is the set of relations.

Definition 2 (Meta-path) A meta-path M of length l is a path on the schema graph T_G which can be defined as $M = t_1 \xrightarrow{r_1} t_2 \xrightarrow{r_2} \dots \xrightarrow{r_{l-1}} t_l$, where $t_i \in T$ represents the entity type while $r_i \in R$ denotes the relation. A path $P = v_1 \xrightarrow{r_1}$

$v_2 \xrightarrow{r_2} \dots \xrightarrow{r_{l-1}} v_l$ is a path instance of meta-path M if $\forall i \in \{1, \dots, l\}, t_i \in \tau(v_i)$ and $\forall i \in \{1, \dots, l-1\}, e_i \in \phi(v_i, v_{i+1})$. In this case, entity pair (v_1, v_l) is connected by an path instance of M , denoted as, $\mathbb{I}_M(v_1, v_l)$

IV. APPROACH

In this section, we will introduce the proposed framework, called LLM4HIN, for mining meta-paths over HINs with LLMs. The overall framework is illustrated in Fig 2, which contains three main components: 1) an LLM-based meta-path generator leverages semantics and structural information to generate meta-paths. 2) a meta-path ranker to estimate the quality of generated meta-paths on HINs, and 3) a meta-path-based logical reasoning module to conduct reasoning over HINs for downstream tasks.

A. LLM-based Meta-path Generator

Conventional strategies for meta-path mining in HINs encounter two pivotal difficulties. Embedding methods, which are representative of structural approaches, frequently fail to adequately capture semantic information [29], [30], necessitating the use of meta-paths as input. Conversely, RL-based approaches are computationally demanding due to the extensive computations and random walks involved [22]. Our methodology, LLM4HIN, mitigates these challenges to a considerable extent. It is fundamentally composed of two principal components: a meta-path sampler and a meta-path generator rooted in LLM.

Meta-path sampler. To enable LLMs to comprehend the internal structure of HINs, we extract instance-level paths, providing essential structural information. These instances can to some extent represent meta-paths at the entity level [31]. Given its systematic and efficient search of all possible path

Meta-path Generation Prompt

"In the context of Heterogeneous Information Networks (HINs), a meta-path is a formalized definition of a composite relation between multi types of nodes in the network. A meta-path can explain a certain relation below are example meta-paths. Each meta-path should start with an type, involve interactions between types and relations, and end with an type. You need to generate meta-paths with {length} words in total."

Meta-path examples are as follows:

```
['politician', 'isAffiliatedTo', 'party', 'isLocatedIn', 'country']
['scientist', 'hasAcademicAdvisor', 'person', 'livesIn', 'country']
```

.....

```
['astronaut', 'isAffiliatedTo', 'party', 'isAffiliatedTo_inv', 'officeholder', 'isPoliticianOf', 'country']
```

Do not give me any explanation or any other information. Just give me {k} more meta-paths to explain relation 'isCitizenOf'."



LLM output

1. ['person', 'isCitizenOf', 'country', 'isPoliticianOf_inv', 'officeholder', 'isCitizenOf', 'country']
2. ['person', 'isCitizenOf', 'country', 'hasNeighbor', 'country', 'hasNeighbor_inv', 'country']
3. ['person', 'isCitizenOf', 'country', 'hasCapital', 'town', 'hasCapital_inv', 'country']

Fig. 3. Example input and output in meta-path generation. The Prompt contains background information, examples, and instructions.

instances between anchors, we utilize Breadth-First Search (BFS) for path instance extraction.

For a given triple (e_1, r, e_k) , a path instance can be represented as a sequence of triples where the head and tail of each triplet are connected in turn, i.e., path instances P can be illustrated as follows: $P = \{(e_1, r_1, e_2), (e_2, r_2, e_3), \dots, (e_{k-1}, r_k, e_k)\}$. Specifically, in Fig 1, a path instance for $(GeorgeWashington, isCitizenOf, America)$ can be represented as $P = \{(GeorgeWashington, BornIn, Virginia), (Virginia, LocatedIn, Amercia)\}$. In the presence of a target relation, our method initially selects a subset of triples that incorporate this relation. Subsequently, for each chosen triple, we conduct a breadth-first search to identify all path instances of the specified length connecting the head entity to the tail entity. Owing to the potential for entities to possess multiple types, it is imperative to identify common types to determine meta-paths. The Lowest Common Ancestor (LCA) algorithm efficiently identifies the minimum common ancestor of two types. Consequently, we employ LCA to derive meta-paths from path instances in this paper.

Meta-path Generator with LLM. After deriving meta-paths for a specific relation, we render them in a natural language construct that facilitates the LLMs' comprehension of the semantic nuances encoded within the meta-paths. For the inverse of an existing relation, we denote it by prefixing the "inv_" symbol.

Although the sampled meta-paths represent a limited subset of the entire spectrum, it remains impractical to input all meta-paths into the LLMs. Consequently, we randomly choose k paths as prototypes to feed into the LLMs, prompting them to generate additional meta-paths. We supply the model with a predefined set of permissible types and relations, thereby ensuring that the model's outputs remain within the framework of sanctioned relationships. Furthermore, post-generation, we

refine the derived rules by pruning any illegitimate meta-paths, thereby arriving at a final suite of meta-paths generated by the LLMs.

B. Meta-path Ranker

Due to the significant hallucination problem in LLM [32], it is necessary to measure the effectiveness of meta-paths. Confidence and coverage metrics intuitively reflect the accuracy and generalizability of meta-paths respectively. Therefore, we employ confidence and coverage metrics to filter the rules produced by the model. Here, we briefly introduce the definitions of confidence and coverage.

Coverage We conceptualize the coverage of M in terms of its ability to identify the query relation r_q as the proportion of entity pairs connected by both r_q and an path instance of M to all entity pairs connected by r_q in HIN $\mathcal{H}$,

$$Cov_{\mathcal{H}}(r_q|M) = \frac{|\{(v_i, v_j) | \mathbb{I}_M(v_i, v_j), r_q \in \phi(v_i, v_j), v_i, v_j \in \mathcal{H}\}|}{|\{(v_i, v_j) | r_q \in \phi(v_i, v_j), v_i, v_j \in \mathcal{H}\}|} \quad (1)$$

Confidence The confidence of meta-path M in identifying r_q given facts in $\mathcal{H}$ is described as the ratio of entity pairs connected by both r_q and a path instance of M to all entity pairs connected by an path instance of M ,

$$Conf_{\mathcal{H}}(r_q|M) = \frac{|\{(v_i, v_j) | \mathbb{I}_M(v_i, v_j), r_q \in \phi(v_i, v_j), v_i, v_j \in \mathcal{H}\}|}{|\{(v_i, v_j) | \mathbb{I}_M(v_i, v_j), v_i, v_j \in \mathcal{H}\}|} \quad (2)$$

Obviously, the higher the confidence and coverage of a meta-path, the broader the scope of entity pairs it can encompass. We set distinct thresholds for each metric and filter meta-paths with better quality accordingly.

C. Meta-path Reasoning Module

Following the ranking of meta-paths, we can leverage established techniques for logical reasoning or to tackle downstream tasks. In our experimental framework, we utilize the

derived meta-paths for link prediction to access their efficacy, evaluating the quality through performance metrics.

Meta-path reasoning involves computing the similarity between two entities using meta-paths. Given an entity pair and the meta-paths that connect them, metrics like the count of meta-paths, binary features, and confidence features are commonly employed. The binary feature is a vector that denotes the connectivity of meta-paths between pairs, whereas the confidence feature replaces the binary feature's 1s with the confidence scores of the respective meta-paths. In light of the meta-path quality, we employ LASSO regression [33], a linear regression model that integrates the last two features, to determine the similarity between entity pairs.

V. EXPERIMENT

A. Experimental settings

To gauge the efficacy and efficiency of LLM4HIN, we perform link prediction tasks employing the generated meta-paths. Link prediction serves as an empirical and quantifiable measure for assessing the utility and predictive capability of the unearthed meta-paths, enabling us to ascertain their utility in the context of network analysis.

In the realm of HINs, link prediction entails the following: Given any pair of nodes within a HIN, the objective is to forecast the presence of an edge of a particular type between them or to estimate the probability of their connectivity through a meta-path.

Datasets We conduct experiments on two real-world online HINs and knowledge bases, Yago [1] and NELL [2], which exhibit more intricate type and relation mappings compared to schema-simple HINs. Three relations for each dataset are evaluated for link prediction. $\{isCitizenOf, DiedIn, GraduatedFrom\}$ in Yago and $\{WorksFor, CompetesWith, PlaysAgainst\}$ in NELL.

TABLE I
STATISTICS OF REAL-WORLD DATASETS

Dataset	Instance Graph			Schema Graph	
	#Entity	#Relation	#Facts	#Entity Types	#Meta-relations
Yago26K-906	26,078	34	390,738	906	30
NELL	49,869	833	296,013	285	827

Baselines We compare LLM4HIN with eight established relation reasoning methods that encompass meta-path-based, embedding-based, and multi-hop-based approaches. These methods are MPDRL, PCRW, Autopath, Metapath2Vec, HIN2VEC, RotatE, TransE and MINERVA. Their implementations and brief introduction are as follows.

- **MPDRL [22]** MPDRL use an RL agent to discover path instances and summarize them into meta-paths¹.
- **PCRW [18]** PCRW mines meta-paths through random walk, the implementation is released with [18].

¹<https://github.com/mxz12119/MPDRL>

- **Autopath [21]** Autopath calculate the similarity between pairs empirically through the probabilities of arrival by a trained agent².
- **Metapath2Vec [30]** Metapath2Vec implement random walk on meta-paths to construct node embeddings³.
- **HIN2Vec [34]** HIN2Vec⁴ exploits different types of relationships among nodes to capture the semantics embedded in HINs
- **RotatE [35]** RotatE⁵ learns embeddings to represent entities and relations in KBs. It can be used in HINs with type information removed.
- **TransE [36]** TransE makes the sum of the head/relation vector close to the tail vector to construct the embeddings⁶.
- **MINERVA [37]** MINERVA is a multi-hop approach for automated reasoning based on neural RL⁷.

Metrics. Two metrics are employed to validate the efficacy of our method: the area under the receiver operating characteristic curve (ROC-AUC) and average precision (AP). Each method is independently executed 5 times, and the metrics are presented in (mean $\pm$ standard deviation).

B. Link Prediction Results

For a certain relation task, e.g. *isCitizenOf*, we use a BFS sampler to obtain some path instances and then summarize them into meta-paths for *isCitizenOf*. Using a LLM-based meta-paths generator, more meta-paths for *isCitizenOf* are generated based on examples and background information, an example prompt is shown in Fig 3. After applying a meta-path ranker using coverage and confidence to all the meta-paths, the ranked meta-paths can be used in link prediction.

The detailed results are presented in the table II. The experimental outcomes reveal that LLM4HIN outperforms all comparative methods in four out of the six relations under investigation. In the YAGO dataset, LLM4HIN demonstrates superiority over all baselines. The average AUC is 0.885, suggesting that the vast majority of triples in the validation set are encompassed by the generated meta-paths. This indicates that the meta-paths are of high precision, with an average AP of 0.926. In NELL, LLM4HIN exhibits notable performance in relation *WorksFor*. Although LLM4HIN does not surpass several baselines in relation *Competewith* and *PlayAgainst*, it still maintains a relatively high level compared to the other baselines.

Moreover, it is noteworthy that LLM4HIN generates meta-paths through a pre-trained model, which substantially reduces the time required for discovery. In contrast, embedding-based approaches necessitate extensive embedding operations, and RL-based models demand prolonged training and learning phases. LLM4HIN can discern meta-paths at a rapid

²<https://github.com/yangji9181/AutoPath>

³<https://ericdongyx.github.io/metapath2vec/m2v.html>

⁴<https://github.com/csiesheep/hin2vec>

⁵<https://github.com/DeepGraphLearning/KnowledgeGraphEmbedding>

⁶<https://github.com/thunlp/OpenKE>

⁷<https://github.com/shehzaadzd/MINERVA>

TABLE II
ROC-AUC AND AP RESULTS FOR YAGO AND NELL, AVERAGED OVER 5 RUNS. THE BOLD/UNDERLINED RESULTS RESPECTIVELY INDICATE THE BEST/SECOND BEST PERFORMANCES FOR EACH RELATION AMONG ALL METHODS.

		LLM4HIN	MPDRL	PCRW	Autopath	Metapath2Vec	HIN2Vec	RotatE	TransE	MINERVA
Yago										
isCitizenOf	ROC-AUC	0.926 ± 0.022	0.781 ± 0.050	0.579 ± 0.036	0.757 ± 0.033	0.652 ± 0.043	0.800 ± 0.067	0.729 ± 0.094	0.590 ± 0.072	<u>0.818 ± 0.045</u>
	AP	0.938 ± 0.018	0.796 ± 0.032	0.716 ± 0.020	0.723 ± 0.059	0.791 ± 0.054	0.834 ± 0.082	0.819 ± 0.085	0.810 ± 0.082	<u>0.840 ± 0.045</u>
DiedIn	ROC-AUC	0.897 ± 0.034	0.710 ± 0.006	0.645 ± 0.020	0.725 ± 0.027	0.661 ± 0.031	<u>0.785 ± 0.027</u>	0.784 ± 0.081	0.622 ± 0.077	0.635 ± 0.040
	AP	0.940 ± 0.026	0.679 ± 0.047	0.683 ± 0.039	0.782 ± 0.034	0.830 ± 0.031	0.877 ± 0.040	<u>0.885 ± 0.086</u>	0.745 ± 0.079	0.786 ± 0.019
GraduatedFrom	ROC-AUC	0.819 ± 0.044	0.664 ± 0.014	0.586 ± 0.008	0.727 ± 0.032	0.665 ± 0.055	<u>0.803 ± 0.050</u>	0.646 ± 0.067	0.664 ± 0.064	0.619 ± 0.016
	AP	0.896 ± 0.039	0.743 ± 0.035	0.662 ± 0.027	0.728 ± 0.061	0.783 ± 0.059	0.847 ± 0.041	<u>0.888 ± 0.044</u>	0.754 ± 0.023	0.718 ± 0.057
NELL										
WorksFor	ROC-AUC	0.910 ± 0.009	0.762 ± 0.034	0.644 ± 0.011	0.703 ± 0.010	0.613 ± 0.012	0.790 ± 0.009	<u>0.866 ± 0.022</u>	0.634 ± 0.034	0.765 ± 0.028
	AP	0.913 ± 0.006	0.873 ± 0.022	0.737 ± 0.044	0.778 ± 0.019	0.821 ± 0.038	0.870 ± 0.020	<u>0.911 ± 0.028</u>	0.717 ± 0.036	0.802 ± 0.017
PlaysAgainst	ROC-AUC	<u>0.927 ± 0.004</u>	0.776 ± 0.030	0.570 ± 0.003	0.542 ± 0.011	0.764 ± 0.033	0.786 ± 0.028	0.967 ± 0.010	0.845 ± 0.020	0.544 ± 0.008
	AP	0.923 ± 0.004	0.823 ± 0.015	0.745 ± 0.005	0.693 ± 0.011	0.913 ± 0.006	0.917 ± 0.008	0.993 ± 0.008	0.928 ± 0.013	0.683 ± 0.025
CompetesWith	ROC-AUC	0.761 ± 0.006	0.601 ± 0.005	0.546 ± 0.007	0.567 ± 0.018	0.600 ± 0.008	0.731 ± 0.025	0.872 ± 0.022	<u>0.773 ± 0.037</u>	0.585 ± 0.021
	AP	0.844 ± 0.010	0.695 ± 0.030	0.701 ± 0.041	0.689 ± 0.026	0.735 ± 0.014	0.863 ± 0.008	0.939 ± 0.018	<u>0.908 ± 0.025</u>	0.709 ± 0.015

pace. Table III illustrates the average time expenditure for LLM4HIN to generate 50 distinct meta-paths. Despite increasing criticisms regarding the propensity of LLMs for repetition [32], the time consumption of LLM4HIN is still more frugal than that of the extant methods for meta-path discovery.

TABLE III
TIME CONSUMPTION FOR LLM4HIN TO GENERATE 50 META-PATHS

Relation	Time(s)	Relation	Time(s)
<i>isCitizenOf</i>	80.144	<i>CompeteWith</i>	183.731
<i>GraduatedFrom</i>	75.532	<i>PlayAgainst</i>	446.235
<i>DiedIn</i>	105.62	<i>WorksFor</i>	536.256

In Table III, we show the time consumption for LLM4HIN to generate 50 meta-paths for each relation. Combined with performance in link prediction, we can conclude that less time-cost means less hallucination, resulting in better results. Next, we move on to analyze meta-paths directly illustrating how LLM4HIN excels most of the baselines.

C. Analysis of Meta-paths

Some meta-paths generated by LLM4HIN are shown in Table IV. It is evident that LLM4HIN produces comprehensive meta-paths. Taking the relation *WorksFor* as an instance, the types {*Chef*, *Athlete*, *CEO*, *Journalist*} are all subtypes of *Person*. Utilizing its formidable natural language comprehension capabilities, LLM4HIN generates meta-paths that are challenging to uncover with embedding or RL-based methods. Furthermore, LLM4HIN discovers these meta-paths with only a few examples, rather than undergoing prolonged learning phases. Consequently, we can deduce that it is the extraordinary exploitation of semantic information that enables LLM4HIN to discern elusive meta-paths, thus outperforming most comparative methods.

Additionally, to further illustrate the efficacy of LLMs within LLM4HIN, we conduct ablation studies.

D. Ablation Experiments

1) *Removing Sampled Metapaths*: We conduct ablation experiments to make a comparison between meta-paths sampled directly from HINs and those generated by LLM4HIN.

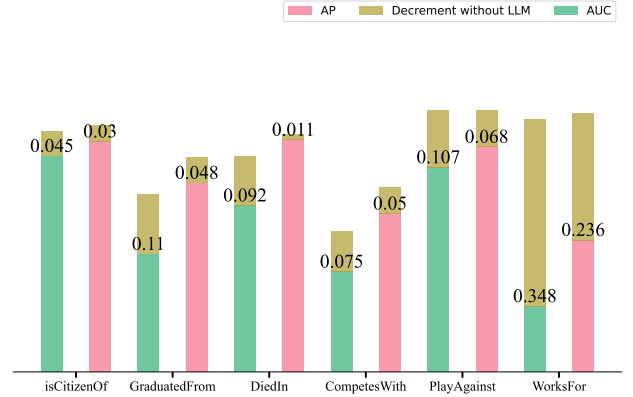


Fig. 4. Results of ablation experiments.

Fig 4 illustrates that the LLM plays a significant role in LLM4HIN. Upon removal of LLM, there is a consistent decrease observed in the AP and AUC values for each relation. Most notably, *WorksFor* experiences the largest drop, diminishing by 34.8%.

2) *Varying Base LLM Model*: We analyze the impact of employing different LLMs by evaluating ChatGLM [38] and Llama2 [39] at scales of 7B, 13B, and 70B, using ROC-AUC to compare their performance in link prediction on the YAGO dataset. As shown in Table VI, LLM4HIN achieves stable performance across different LLMs, demonstrating its robustness to LLM selection. The results reveal that larger models do not always yield better outcomes. Furthermore, due to computational resource constraints, large-scale models are often subject to truncation and early stopping issues.

TABLE IV
EXAMPLE META-PATHS FOUND BY LLM4HIN

Relations	Meta-path
isCitizenOf	Person $\xrightarrow{\text{BornIn}}$ Administrative $\xrightarrow{\text{LocatedIn}}$ Country
	Economist $\xrightarrow{\text{LivesIn}}$ District $\xrightarrow{\text{LocatedIn}}$ Country
	Scientist $\xrightarrow{\text{Influences}}$ Economist $\xrightarrow{\text{LivesIn}}$ Country
	Person $\xrightarrow{\text{LivesIn}}$ Country
	Officeholder $\xrightarrow{\text{WorksAt}}$ University $\xrightarrow{\text{LocatedIn}}$ Country
GraduatedFrom	Officeholder $\xrightarrow{\text{WorksAt}}$ University
	Person $\xrightarrow{\text{WorksAt}}$ University
	President $\xrightarrow{\text{DiedIn}}$ Town $\xrightarrow{\text{IsLocatedIn}^{-1}}$ University
	Person $\xrightarrow{\text{HasAdvisor}}$ Scientist $\xrightarrow{\text{WorksAt}}$ University
	Scientist $\xrightarrow{\text{MarriedTo}^{-1}}$ Scientist $\xrightarrow{\text{WorksAt}}$ University
WorksFor	Person $\xrightarrow{\text{BelongsTo}}$ Company
	Chef $\xrightarrow{\text{BelongsTo}}$ Company
	Athlete $\xrightarrow{\text{ControlledBy}}$ School
	CEO $\xrightarrow{\text{BelongsTo}}$ Company
	Journalist $\xrightarrow{\text{WritesFor}}$ Company
PlayAgainst	Team $\xrightarrow{\text{CompetesWith}}$ Team
	Team $\xrightarrow{\text{CompetesWith}}$ Coach $\xrightarrow{\text{HomeStadium}}$ Attraction
	Team $\xrightarrow{\text{InGame}}$ Game $\xrightarrow{\text{Participant}}$ Team
	Coach $\xrightarrow{\text{BelongsTo}}$ League $\xrightarrow{\text{LeagueTeam}}$ Team
	Team $\xrightarrow{\text{SubpartOf}}$ League $\xrightarrow{\text{LeagueTeam}}$ Team

TABLE V
AVERAGE DECLINE WITHOUT LLM

Average Decline	AP	AUC
Yago	2.952%	8.218%
NELL	11.818%	17.642%

TABLE VI
RESULTS OF DIFFERENT LLMs, WHERE BOLD AND UNDERLINED VALUES
DENOTE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY.

	isCitizenOf	DiedIn	GraduatedFrom
ChatGLM	<u>0.919</u>	0.881	0.800
Llama2-chat-7B	0.917	<u>0.886</u>	0.813
Llama2-chat-13B	0.927	0.897	<u>0.819</u>
Llama2-chat-70B	0.913	0.891	0.823

E. Inductive Experiments

Overall, RotatE consistently demonstrates superior performance across all metrics and outperforms LLM4HIN in *CompeteWith*. To evaluate the generalization capabilities of LLM4HIN, we conduct inductive experiments using the NELL dataset. We sample 40% of the positive testing set and remove all nodes that appeared in the pairs from the instance graph. We employ the same metrics as previously. The results are presented in Fig 5.

As depicted in Fig 5, the AUC of RotatE decreases to 0.558, whereas that of LLM4HIN persists at 0.818. Regarding AP, RotatE falls to 0.770, while the LLM4HIN maintains 0.866. In summary, LLM4HIN exhibits superior generalization capabilities compared to RotatE. Moreover, when compared to

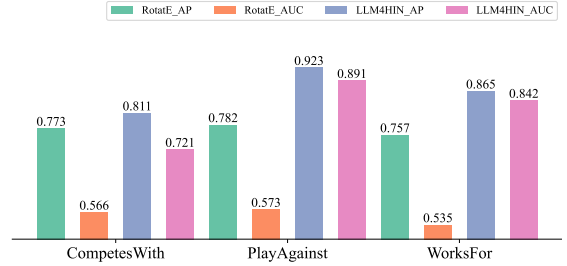


Fig. 5. Results of inductive experiments.

other baselines, it is evident that even without the removal of nodes, several baselines still fall short of LLM4HIN's performance, further highlighting the robustness and stability of LLM4HIN.

VI. CONCLUSION

In this paper, we harness the advanced natural language processing capabilities of LLMs to introduce LLM4HIN for the extraction of meta-paths in HINs. LLM4HIN initiates by selecting a subset of meta-paths through sampling, subsequently utilizing LLMs to proliferate these meta-paths. Following validation, these meta-paths are employed in link prediction tasks, yielding commendable outcomes. Moreover, through ablation studies, we elucidate the critical role of semantic comprehension in LLM4HIN's efficacy in meta-path discovery. Subsequently, we conduct inductive experiments by contrasting LLM4HIN with RotatE, the preeminent baseline model, to corroborate its superior generalizability.

Our ongoing research endeavors will concentrate on refining LLMs' comprehension of knowledge within HINs, thereby enhancing their proficiency in logical reasoning based on meta-paths within HINs.

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